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Points to Consider No. 15

Mobile Manufacturing

2025

PDA Parenteral Drug Association

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Introduction

The manufacturing of drugs has historically been fixed in the building in which it is processed and therefore, manufacturing was conceptualized as fixed in nature. Specifically, the conventional construction of such manufacturing facilities has been executed on the site and viewed as immobile. Such construction has often been termed stick-built or been called brick-and-mortar facilities. This type of construction has been the dominant method in our industry throughout the past. However, the COVID-19 pandemic highlighted the need for new thinking and a more flexible, modern, and scientifically sound approach to the platform design of the manufacturing process, thereby ensuring the essential manufacture, distribution, and availability of medicinal products. While some concepts like modular facility construction have been implemented, only recently has a significant advancement in technologies allowed alternative methods, such as mobile manufacturing facilities, to be used in its place. Mobile manufacturing is the ability to relocate manufacturing while maintaining an environment suitable for good manufacturing practice (GMP) production. This paper focuses exclusively on biotech and pharmaceutical manufacturing. The environment under consideration is inclusive of the process equipment and cleanrooms enveloping the process that provide the right conditions to manufacture good-quality medicinal products. The extension or boundary limits of this environment will be further considered in the scope of this analysis.

The use of self-contained, mobile manufacturing units enables the relocation or replication of a manufacturing facility between different geographic locations, which allows manufacturers to rapidly supply and scale up product manufacturing. Examples of manufacturing strategic improvements over time abound in the pharmaceutical industry, including the adoption of cleanrooms (first introduced in 1962), which are now a requirement for GMP manufacturing. Prior to that time, it was difficult to control the manufacturing environment, and risks from particulates and adventitious agents were common. Since then, cleanrooms have become the status quo and are a requirement for GMP manufacturing. As each change in pharmaceutical manufacturing occurs, details for expectations have evolved. For cleanrooms, various types of materials have been utilized, with gypsum boards painted with epoxy being one of the most common early options due to gypsum board being so common in the construction industry.

The development of other technologies which evolved from the adoption of cleanrooms presaged mobile manufacturing, such as the introduction of prefabricated elements. The preassembled, prefabricated solutions which altered how manufacturing facilities were built on-site also seeded the concept of manufacturing mobility as it became common to ship these semi-completed facilities across the world. At first, mobile or a commissioned facility was not a consideration as the initial prefabricated technologies became fixed in nature. But within the past 20 years, prefabricated technologies have been introduced that can be easily prequalified, disassembled, and relocated elsewhere. These are the technologies that will be further evaluated within this document to assess the requirements, risks, and advantages within the scope of mobile manufacturing.

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The process equipment used to manufacture pharmaceuticals is prefabricated and prequalified (via factory acceptance testing) off-site prior to delivery at the facility, which lends support to the overall concept of mobile manufacturing. Another major advancement in the evolution of mobile manufacturing feasibility was the introduction in the 1970s of single-use components, which have matured dramatically since that time and provided greater flexibility in facility design. Accompanying the advances in technology, the size of manufacturing facilities and equipment has decreased over time, driven both by process intensification and personalized medicines that target smaller patient populations.

Geopolitical concerns have also been a driver of distributed manufacturing, as many countries look for in-country manufacturing. Similarly, pharmaceutical and biotechnical companies want to mitigate the risk to their assets by having the ability to move their production capabilities in the event of wars, natural disasters, or other crises. Lastly, regulatory barriers within the drug production landscape have also lessened over time to accommodate the concept of manufacturing mobility.

Scope

This document aims to assess the feasibility and requirements for designing and qualifying a self-contained manufacturing process that, once validated, can be relocated and operated in a different location for the same process, without the requirement for regulatory preapproval or revalidation.

The scope includes exploring the technology supporting mobile manufacturing in greater detail and the regulatory requirements and risks associated with moving a manufacturing process from one location to another. While mobile manufacturing can be considered an enabling technology for distributed manufacturing or point-of-care manufacturing, the scope of this paper excludes identifying or elaborating on those requirements.

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Glossary

Definitions have been provided to help clarify the concepts discussed in this document. While some definitions used vary among companies, geographic location, etc., the definitions described below are consistently used within this Points to Consider document. Where a definition is based on another published source, the source is cited.

Distributed Manufacturing (DM)
A decentralized manufacturing strategy consisting of a manufacturing platform comprising manufacturing units deployed to multiple locations. Possible use scenarios include:

- Units located within manufacturing facilities operating within the host's pharmaceutical quality system (PQS).
- Units manufactured and installed to the same specifications at multiple manufacturing facilities, networked and operated by a central remote PQS.
- Units as independent manufacturing facilities, each with its own PQS ***(2)***.

Family approach
A concept applied to equipment qualification where a set of identical or very similar pieces of equipment (i.e., those deemed functionally equivalent) are subjected to reduced qualification after an initial member of the set is fully qualified.

Initially qualified site
The first site where a mobile unit is qualified for producing a specific pharmaceutical product or set of products prior to deployment to other locations (usually a fixed facility of the pharma company that developed the product to be manufactured by the mobile units).

Mobile Manufacturing
The manufacturing of pharmaceutical products at off-site locations via qualified manufacturing units.

Mobile Manufacturing Unit (MMU)
A deployable manufacturing unit that has been qualified to make a specific set of pharmaceutical products, regardless of location. It is designed so that it can be easily disassembled and reassembled into its functional state without incorporating any major rework.

MMU vendor
The company that assembles the mobile manufacturing units and provides them to the initially qualified site. These are usually enabling companies that provide units, modules, and associated services to end-user pharmaceutical companies and perform the initial qualification of major facility systems like water and air, but not product- or process-specific aspects of validation.

Off-site
The site where the MMU is deployed for product manufacture.

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On-site
The site where the MMU is fully qualified (initially qualified site) for manufacture of a pharmaceutical product or set of products.

Note: This is usually a fixed site at a pharma company.

Abbreviations

FEI	FDA Establishment Identification
MMU	Mobile Manufacturing Unit

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Topic 1. Technology Considerations

Q 1.1. What are the available options to build a facility?

The means to fabricate or build a good manufacturing practices (GMP)-compliant facility can be categorized into two main groups: on-site construction and prefabricated solutions.

One method for on-site construction involves conventional stick-built or brick-and-mortar facilities. This involves various trades (e.g., carpenters, electricians, masons, etc.) being on-site and using stock materials that are cut or manipulated on-site based on the build plans for the facility. Another approach to on-site construction is the use of modular components, where the materials are built off-site but assembled on-site based on design plans. Both options provide a fixed facility which cannot be moved to an alternate location if needed.

Prefabricated solutions allow for the build-out of completed components assembled off-site. Being built off-site entails certain aspects of mobility, and the level of completeness or percentage of work done off-site typically lends itself to the ability to redeploy the facility elsewhere. However, there are certain prefabricated solutions that, once installed, become fixed in place or not easily relocated. To be considered a mobile manufacturing platform, the ease of redeployment becomes a prerequisite. To distinguish, the two prefabricated units, which are inherently mobile, will be referred to as mobile manufacturing units (MMUs).

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Q 1.2. What are the advantages and challenges associated with building a mobile manufacturing unit?

There are several potential advantages of MMUs that include:

- **Drug supply robustness:** Enables a rapid-response and mobile or localized production capability and availability of drugs could fill critical supply-chain gaps.
- **Rapid response capabilities:** A platform approach that enables deployment of new formulations, raw materials, and processes to the MMU which could enable a more agile response to man-made and natural disasters, as well as alleviate last-mile distribution challenges.
- **Customization and personalized medicine:** This type of manufacturing technology also has the potential to benefit patients though smaller-batch, customized formulations that employ emerging technologies in precision medicines by leveraging pharmacogenomics and artificial-intelligence platforms.
- **Agile manufacturing networks:** Allows a company to flex their manufacturing capacity in capacity or geographically without the overhead costs of traditional facilities.
- **Creation of new business models:** From a business perspective, mobile manufacturing supports innovation and provides a pathway for new technologies and the emergence of new business models that align with patient outcomes.

Challenges of MMUs could be associated with the following:

- **Scale:** As MMUs need to be prefabricated and transported, very large operations would be prohibitive. For example, installations involving large 10,000 L stainless-steel bioreactors would not be feasible.
- **Weight:** Depending on the floor structure, it would be challenging to move the unit with extremely high-point loading equipment installed.
- **Structural requirements:** Different countries have different structural requirements depending on seismic activity or potential weather events (e.g., hurricanes). If the correct design space was not considered in advance, moving to a new location may not be possible.
- **Utility requirements:** Movement to a new country may encounter local code constraints. Utility requirements, such as portable power supply and water generation, need to be considered in advance to simplify the approach or enable retrofitting based on the new location.
- **Handling of process equipment:** Determination of how the shipment of process equipment will be managed in the event of redeployment (e.g., shipped inside MMU versus re-crated and shipped separately).

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Q 1.3. How does one implement and validate a mobile manufacturing unit?

At the core of an MMU is the process technology, that is, the equipment being used to produce either the drug substance and/or drug product. Working backwards from there is the design space of the MMU, which includes the cleanrooms, building automation system/building management system, qualified building automation system/qualified building management system/environmental monitoring system, process automation system, and mechanical electrical piping to operate said area. It is important that all these elements reside within the MMU, so that when it is redeployed, those elements stay with the system. Such MMUs could potentially be placed outdoors or within a host facility that would provide basic inputs or utilities.

Establishing a proper design space for the MMU while simultaneously maintaining the ease of mobility for all aspects of the unit is critical. Typically, multiple vendors (e.g., process equipment, cleanroom, utility suppliers) contribute to what becomes the final MMU, and all parties working together from the start is critical to success since the solutions are built off-site at different locations.

Additionally, establishing clear user requirements for MMU mobility is critical to redeployment. For example, if the overall design space of the MMU needs to account for various locations in a country or around the world for redeployment, the MMU design space needs to consider a worst-case scenario or incorporate features that allow the MMU be retrofittable to a different location in the future.

Other geographical considerations include local code requirements where the unit could be redeployed. If considering multiple countries for relocation, due diligence must be undertaken for electrical systems, or example, since the voltages available may vary across countries and regions. Structural considerations also need to be considered as different countries have varied regulations and different seismic activity or potential weather events (e.g., hurricanes).

MMU requirements depend on the process and utilities needed to support the MMU infrastructure. Consideration of host facilities or locations within the manufacturing network that could house said MMU should be conducted during a preliminary evaluation phase. Based on required utilities of the MMU and the availability of those utilities within the network, a design space needs to be defined which includes utility generation within the scope of the MMU, if deemed necessary. If required utilities are available at all or most locations where the MMU may be deployed, this would negate the need to include that component within the MMU design space.

Certain aspects of commissioning, qualification, and validation (CQV) can take place at the vendor's site, typically known as factory acceptance tests (FAT), with additional testing, referred to as site acceptance tests (SAT), taking place at the final location. Leveraging the FAT/SAT into the CQV or master validation strategy will help mitigate redundant testing during qualification.

Elements to consider for validation of an MMU:

- Identify the need for an MMU
- Identify the initially qualified site
- Determine what unit operations will be performed at the MMU

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- List the systems that need qualification and validation, which include but are not limited to:
  - Utilities
  - Manufacturing systems
  - Computer systems
  - Servers
  - Access control
  - Unit operation validation
  - Method validation
  - Cleaning validation
  - Shipping validation
- Describe procurement strategy
- Identify training requirements
- Define quality oversight

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Q 1.4. How can mobile manufacturing facilitate technology transfer?

Mobile manufacturing has the potential to facilitate transfer of manufacturing and/or analytics geographically, once an MMU is established. The likely order of establishment of capabilities would first be the development and qualification of manufacturing or analytics at an initially qualified site. The initially qualified site could be fixed in nature or an MMU. Next, once the specific capabilities are established, they can then be tech-transferred to the subsequent MMU, and qualification activity for the specific capabilities would take place on the MMUs as necessary. The capabilities can be tech-transferred and qualified in more than one MMU, and the process can be streamlined for subsequent units from the initial transfer to the first unit.

Question 1.3 addressed how to establish the design space of the MMU in conjunction with the expectations of the host facility. Manufacturing and analytics, raw materials, reagents, and disposables may need to be qualified, probably at the initially qualified site, before being loaded onto the MMU(s) and the MMUs being deployed.

Implementation of a network of MMUs that are equivalent in design and operation could potentially reduce post-marketing regulatory submissions associated with tech-transfer. An intended use of some types of mobile-manufacturing technologies may be rapid scale-out and/or deployment to locations in close proximity to patients. These types of intended uses (e.g., leveraging tech-transfer among a network of MMUs) could potentially be validated during product development such that data to support these activities could be provided in the original marketing application. Tech-transfer criteria could then be established that ensure process performance consistency and comparable product quality that, if met, could reduce post-marketing regulatory submissions.

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Topic 2. Regulatory Considerations

Q 2.1. How do current regulations impact the adoption of mobile manufacturing?

Current regulatory expectations with respect to facilities used in the manufacturing of drugs and biologics is that they are required to be designed in compliance with cGMPs. At a high level, processing should permit a smooth flow of materials through the facility. Space must be provided for receiving, quarantine, sampling, and storage of all manufacturing materials, including intermediate and finished bulk drug substance.

More specifically, facility builds or modifications need to ensure compliance with a host of regulations, such as the U.S. Food and Drug Administration (FDA) Title 21, Code of Federal Regulations, the European Medicines Agency (EMA) Commission Decision 2021/1240 of 13 July 2021, and the equivalent requirements of other global regulators. **Appendix A** provides examples of applicable compliance expectations.

As a requirement of licensure, marketing authorization holders (MAHs) must provide information on the manufacturing locations, such as the address (and potentially global positioning system (GPS) coordinates), that will allow the relevant regulatory agencies to identify the manufacturing location. The location needs to hold a manufacturing license and a GMP certificate. In some jurisdictions, the facility locations and specific GMP activities (e.g., analytical testing, storage, manufacturing) must be registered and assigned specific identification and status by the regulators. For example, the FDA assigns a Firm Establishment Identifier (FEI) number to all facilities, and the identification of the facility or site is linked to an address and specific business entity (e.g., DUN & Bradstreet number). If multiple locations are listed in the dossier (i.e., the application), then this is required for each location. The requirements to file and register facility-specific information, including addresses and site contacts, are in large part linked to the regulators, enabling them to triage inspections.

In an ideal case of mobile manufacturing, the MMU could move from one physical location to another, and the license and certificate would only be required once. However, there are no provisions in current regulations for such a scenario. Moreover, in the United States, many state boards of pharmacy require registration for in-state drug manufacturers, presenting additional complexity of rapidly moving across state lines. In addition, the movement of the MMU could create logistical challenges for regulators when making decisions about where and when they would need to deploy resources to conduct GMP inspections. Also, the full manufacturing process may be segmented in different locations (e.g., warehousing and the quality control of raw materials may be centrally located for stocking of one or a set of MMUs).

Finally, some countries have multiple agencies that would first need to agree to accept the license and certificate from another agency within their country. Typically, countries will not have provisions to accept such licenses from other agencies. The situation becomes even more convoluted and complicated if the MMU is expected to cross country borders ***(1)***.

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Q 2.2. What actions can global regulators take to help facilitate the implementation of mobile manufacturing?

Regulators can facilitate mobile manufacturing both by adjusting regulatory/administrative procedures and modifying validation and qualification expectations.

A prime example of where regulatory requirements could be revisited is the FDA approach to the FEI number. It would not be practical to give each MMU an FEI number: there would be too many units in circulation and the MMU manufacturer would need to reapply for a new one each time an MMU relocated, potentially every month. Not only would it be burdensome for the manufacturer but for FDA, with little public health benefit. Alternatively, an approach could be to have an FEI number for the site where the MMUs are produced and then have that site's Quality Unit track the unit numbers from there. This would ensure the traceability expected by regulators, and regulators could revisit the concept of 100% "facility" inspection. If the MMU was viewed as analogous with a "facility," as set forth in 21 CFR 201.1, for example, it would be impractical. A strict interpretation of this regulation may require that each and every mobile unit be inspected after they are produced ***(2)***.

Novel validation and qualification approaches would also ease otherwise insurmountable regulatory burdens. For example, implementing a "family approach" towards validation and qualification of MMUs would streamline mobile unit qualification, without lowering quality standards. In this approach, once one unit is qualified and validated, identical follow-on units would be subject to reduced testing. As another streamlining concept, the bulk release of raw materials or water could take place at one mobile manufacturing site, with portions of the pre-released material distributed to each unit. This would avoid unit-specific testing each time an MMU is made and stocked with material.

Recent draft guidances issued by the FDA on the Platform Technology Designation Program and the Advanced Manufacturing Technologies Designation Program potentially offer a new opportunity for MMUs ***(3, 4)***. Specifically, these draft guidances propose that once a platform technology has been designated as such, it may be leveraged for other new or future products from the same sponsor or if a full right of reference for the same platform technology is provided to a third party. As currently written, these draft guidances are limited to technology incorporated into a drug or biological product which is essential to its structure or function. Expansion of the guidances, or additional guidance to include the methods of manufacturing and the production platforms themselves, could prove useful to accelerate this paradigm. The designation of a platform manufacturing technology should be considered early in development during the clinical phase of drug development. Other regulators, including Health Canada and the Medicines and Healthcare products Regulatory Agency (MHRA), are creating frameworks for advanced therapeutic products that take into account the scientific and technological advances that are leading to distributed and point-of-care manufacturing, both of which could take advantage of mobile manufacturing facilities ***(5)***.

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Topic 3. Implementation

Q 3.1. How would one validate a mobile manufacturing unit at a new location?

Assuming the initial validation was fully conducted at the initially qualified site, validation of an MMU should be done in stages, with as many elements completed at the initially qualified site as possible (see **Question 1.3**). The initial stage occurs at the manufacturing site of the MMU (vendor), where critical systems like water and air are qualified via installation qualification/operational qualification (IQ/OQ). Mobility over various terrains would be part of this exercise as well. Some systems could be tested at the initially qualified site of the receiving company (i.e., the drug manufacturer). For example, testing and qualification of raw materials should occur at the receiving company's fixed site, as could the initial process validation. These systems would need to be further qualified in field testing, through performance qualification (PQ) or confirmatory runs and, potentially, process validation. This is analogous to "static" and "dynamic" testing of air systems where use in the field would be analogous to "dynamic" testing. In the case of process validation (PV), if the MMU is intended for multiproduct use, it may be challenging to perform PV for every potential product. The choice of the model process for PV should consider its complexity and robustness in order to cover a range of product types.

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Appendix A: Regulatory Compliance Expectations

The following list provides examples of expectations regarding regulatory compliance for mobile manufacturing units (MMU) as per U.S. Food and Drug Administration's 21 CFR Parts 211 and 610 ***(6) (7) (8)***, the European Medicines Agency's Commission Decision 2021/1240 of 13 July 2021 ***(9)***, and the Pharmaceutical Inspection Convention/Cooperation Scheme's (PIC/S) GMP publications ***(10)***. This list is not comprehensive and other rules and regulations may apply. These requirements have historically been established for brick-and-mortar (standing address) facilities and may merit new thinking about how MMUs specifically should comply.

- Mobile units are suitably sized and of appropriate construction to facilitate operations and are adequately designed to prevent contamination, cross-contamination, and mix-ups.
- All utilities (including plumbing and sanitation) are, or will be, appropriately validated, and HVAC systems provide adequate control over air pressure, microorganisms, dust, humidity, and temperature, and sufficient protection or containment as needed.
- Facility, equipment cleaning, and maintenance processes are, or will be, developed and validated.
- Manufacturing equipment are (or will be) qualified and sterile filtration and sterilization processes validated.
- A Quality Unit is established and has the responsibility for oversight of manufacturing, and review and release of components, containers, and closures, labeling, in-process material, and final products.
- Premises and manufacturing equipment are located, designed, constructed, adapted and maintained to suit the intended operations.
- Premises and equipment to be used for manufacturing operations, which are critical to the quality of the products, shall be subject to appropriate qualification and validation.
- Premises and equipment must be located, designed, constructed, adapted, and maintained to suit the operations to be carried out. Their layout and design must aim to minimize the risk of errors and permit effective cleaning and maintenance in order to avoid cross-contamination, build-up of dust or dirt and, in general, any adverse effect on the quality of products.
- Cross-contamination should be prevented for all products by appropriate design and operation of manufacturing facilities. The measures to prevent cross-contamination should be commensurate with the risks. Quality risk management principles should be used to assess and control the risks.

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